

Universal Plastic Shredder V2.0

Community Partner: MME/Ichor Systems

Contact Name: Robert Paxton

Contact email: paxton3@pdx.edu

Contact phone: Optional

IP and Compliance Requirements

- Students must sign an NDA: No
- Intellectual Property will be owned by: Sponsor
- Students must be US persons: No
- Other legal requirements: None

Background and Motivation

During 2023/24, MME and EE capstone students work together to develop a small plastic shredder that reduced common household items into small shreds through the application of fundamental engineering principles. One of the students involved in this project is now wanting to create a larger version of this shredder for use by her company.

From the client – “Semi-conductor equipment manufacturer that produces different types of plastic waste (LDPE, polypropylene, PVC). We are looking to prototype a shredding tool (plastic shredder) that will allow us to recycle and reuse our excess/extra plastic waste and turn the material into other things.”

Objective

Initially, a thorough design review of the original shredder design to determine options for scalability will need to be conducted. At this point, certain aspects of the machine may not scale and will thus need to be redesigned. For example, the original shredder prototype used a single 2HP motor; this prototype will likely need more than one motor and also be significantly larger. This has implications in terms of weight, space occupied, mounting, electrical requirements, shielding etc.

Requirements

TBD by the capstone team based on information from the sponsor.

Suggested Skills

Familiarity with electric motors, industrial control systems.

Deliverables

Working shredding machine that follows industrial wiring and safety standards.